02 / Urological infections

01 Overview and asymptomatic bacteriuria

CLASSIFY
Localised / systemic

CULTURE
Identify the pathogen

REVIEW
Narrow / stop / source control

READ FIRST Adult educational reference. Every dose requires an individual prescription check: indication, allergy, renal/hepatic function, pregnancy, interactions, weight, organism susceptibility and local formulary. Clinical and pharmacy review pending.

HOW TO READ Regimen tables contain drug facts, not independent strength ratings. Strong/Weak applies only when explicitly attached to a recommendation. Orange VERIFY labels identify unresolved source issues; a listed alternative is not interchangeable with first-line therapy.

FORMAT PO = oral; IV = intravenous; IM = intramuscular; TMP-SMX = trimethoprim-sulfamethoxazole. Frequencies are written in full. Single dose means one administration, not a daily course.

Navigation / topic coverage

Antimicrobial / option	Dose and route	Course	Selection / conditions
Lower urinary tract	02 Cystitis; 03 recurrent cystitis	pp 2-3	Treatment and prevention are separate.
Upper / systemic UTI	04-06 Pyelonephritis; 07 systemic UTI	pp 4-7	IV, oral and resistant-organism options.
Devices / sepsis	08 CA-UTI and urosepsis	p 8	Device management and sepsis source-control context.
Male genital / STI	09-10 Urethritis; 11 prostatitis; 12 epididymitis	pp 9-12	Empirical and organism-directed regimens.
Other infections	13 Fournier; 14 HSV/HPV; 15 TB; 16-17 fungal UTI	pp 13-17	Includes non-antibacterial therapies where relevant.
Procedure prevention	18 Periprocedural prophylaxis	p 18	Not treatment of an established infection.

Asymptomatic bacteriuria (ABU): when a treatment table is appropriate

Antimicrobial / option	Dose and route	Course	Selection / conditions
Pregnancy	Culture-directed short course; fosfomycin trometamol is a single-dose option	Agent-specific	EAU Weak. Pregnancy suitability must be checked; see pp 2-3.
Before mucosa-breaching urological procedure	Culture-directed antimicrobial	Procedure-specific	EAU Strong. Do not invent one empirical ABU regimen for all procedures.
No qualifying indication	No antibacterial treatment	None	Includes routine catheter colonisation, renal transplant recipients and recurrent-UTI patients without symptoms; see full 3.3 for population-specific detail.
Immunocompromised / severely ill	Individual assessment	By case	No universal ABU treatment course supplied by EAU.

02 / Urological infections

02 Cystitis: women and men

LOCALISE
Bladder symptoms

SELECT
First-line or conditional alternative

EXCLUDE
Systemic / prostatic involvement

FIRST-LINE Women: fosfomycin, nitrofurantoin, pivmecillinam or nitroxoline (EAU Strong). Aminopenicillins and fluoroquinolones are not routine cystitis choices (EAU Strong).

MEN The 2026 narrative supports pivmecillinam/nitrofurantoin in selected younger men. Prostatic involvement must be reliably excluded for nitrofurantoin; EAU supplies no evidence-based male nitrofurantoin duration. Do not apply the female 5-day course automatically.

Women: oral treatment options

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fosfomycin trometamol	3 g PO once	Single dose	First-line.
Nitrofurantoin macrocrystals	50-100 mg PO four times daily	5 days	First-line; formulation matters.
Nitrofurantoin monohydrate / macrocrystals	100 mg PO twice daily	5 days	First-line.
Nitrofurantoin prolonged-release macrocrystals	100 mg PO twice daily	5 days	First-line.
Pivmecillinam	400 mg PO three times daily	3-5 days	First-line.
Nitroxoline	250 mg PO three times daily	5 days	First-line, where available.
Cefadroxil (or comparable oral cephalosporin)	500 mg PO twice daily	3 days	Alternative; use selectively, considering ecological effects.
Trimethoprim	200 mg PO twice daily	5 days	E. coli local resistance <20%; avoid first pregnancy trimester.
TMP-SMX	160/800 mg PO twice daily	3 days	E. coli local resistance <20%; avoid last pregnancy trimester.

Men: explicit Table 3 course and narrative options

Antimicrobial / option	Dose and route	Course	Selection / conditions
TMP-SMX	160/800 mg PO twice daily	7 days	Explicit male regimen in Table 3; susceptibility and clinical site matter.
Pivmecillinam / nitrofurantoin	See formulation doses above	Verify male course	Narrative options for selected younger men; no separate male pivmecillinam schedule or evidence-based male nitrofurantoin duration is supplied.

02 / Urological infections

03 Recurrence, pregnancy and treatment boundaries

CONFIRM
Culture recurrent episodes

PREVENT
Non-antibiotic measures first

SEPARATE
Acute course vs prophylaxis

PREVENTION The doses below are prophylaxis doses, not treatment of an acute infection. Continuous or postcoital strategies follow discussion of non-antibiotic prevention. Trial durations vary (3-12 months); EAU does not fix one optimal duration.

PREGNANCY Use obstetric/local prescribing guidance. Nitrofurantoin: avoid G6PD deficiency and end of pregnancy. Trimethoprim: avoid first trimester. Sulphonamides: avoid last trimester.

RENAL CHECK Nitrofurantoin is contraindicated at eGFR <30 mL/min/1.73 m2. Drug-specific renal adjustment and monitoring remain required for other agents.

Recurrent cystitis: continuous low-dose prevention

Antimicrobial / option	Dose and route	Course	Selection / conditions
Nitrofurantoin	50 or 100 mg PO once daily	By case	Prophylaxis; formulation, renal function and long-term toxicity review.
Fosfomycin trometamol	3 g PO once weekly	By case	Weekly prevention schedule, not daily therapy.
Trimethoprim	100 mg PO once daily	By case	Choose using local resistance/susceptibility.
Cefalexin (pregnancy option)	125 or 250 mg PO once daily	By case	Pregnancy prevention option in EAU narrative.
Cefaclor (pregnancy option)	250 mg PO once daily	By case	Pregnancy prevention option in EAU narrative.
Postcoital prophylaxis	Agent/dose not separately specified	By case	No distinct postcoital dose is supplied in this EAU section; use a locally agreed regimen.
Self-start acute treatment	Use an appropriate full acute-treatment course	Agent-specific	For selected reliable patients; not the low-dose schedules above.

When the presentation changes

Antimicrobial / option	Dose and route	Course	Selection / conditions
Persistent / early recurrent cystitis	Culture + susceptibility; select another agent	Consider 7 days	For non-resolution or recurrence within 2 weeks; do not simply repeat the failed agent.
Fever / flank pain / systemic illness	Move to systemic-UTI assessment	Site-specific	Bladder-only agents do not substitute for a pyelonephritis regimen.
Pregnancy with symptomatic cystitis	Pregnancy-compatible agent selected by culture and local guidance	Agent-specific	Penicillins, cephalosporins, fosfomycin and selected other agents are discussed; no universal pregnancy dose table is supplied.

02 / Urological infections

04 Pyelonephritis: oral therapy

CULTURE
Urine + susceptibility

IMAGE
Exclude urgent obstruction

REVIEW
Response within 48-72 h

ELIGIBILITY Oral therapy requires an appropriate clinical setting and an agent active at the infection site. Admission, IV therapy and source control depend on illness severity and obstruction.

DO NOT USE Nitrofurantoin, oral fosfomycin and pivmecillinam are not pyelonephritis treatments (EAU Strong).

DURATION CONTEXT These are Table 5 courses. They are not interchangeable with the generic systemic-UTI durations on p 7. EAU advises at least 2 weeks for men with febrile UTI/pyelonephritis or recurrent infection where prostatic involvement is a concern.

EAU Table 5: oral regimen facts

Antimicrobial / option	Dose and route	Course	Selection / conditions
Ciprofloxacin	500-750 mg PO twice daily	7 days	Empirical use requires local fluoroquinolone resistance <10%; see systemic-UTI exclusions on p 7.
Levofloxacin	Standard: 500 mg PO once daily; high dose: 500 mg PO twice daily	5 days	Both dose levels appear in Table 5; choose using local prescribing criteria and renal function.
TMP-SMX	160/800 mg PO twice daily	14 days	Prefer confirmed susceptibility. If used before susceptibility is known, give an initial long-acting parenteral dose (e.g. ceftriaxone).
Cefpodoxime	200 mg PO twice daily	10 days	Oral beta-lactam; susceptibility-guided where possible. If empirical without results, initial long-acting parenteral coverage is advised in the narrative.
Ceftibuten	400 mg PO once daily	10 days	Same empirical beta-lactam caveat; do not assume all oral cephalosporins are equivalent.

Review and transition

Antimicrobial / option	Dose and route	Course	Selection / conditions
Initially IV, then improving	Switch to an active oral agent when oral intake is possible	Total course	Count both IV and oral days; clinical site, susceptibility and response determine the final course.
No improvement / deterioration	Reassess culture, obstruction, abscess and alternative diagnosis	48-72 h or earlier	Persistent illness needs reassessment, not an automatic extension of the same drug.
Pregnancy	Appropriate parenteral therapy; consider oral switch after improvement	7-10 days total	EAU narrative; severe cases require hospital care and obstetric coordination.

02 / Urological infections

05 Pyelonephritis: initial IV options

START
Appropriate IV coverage

CHECK
Renal dose + toxicity

STEP DOWN
Culture-supported oral agent

DOSE BASIS Adult Table 6 doses are displayed without renal adjustment. Standard/high-dose distinctions are preserved. A high-dose entry is not an instruction to use that dose for every patient.

DURATION Table 6 gives daily dosing, not a fixed course for each IV agent. Select the total duration by syndrome, organism, response and oral step-down; see pp 4 and 7.

SELECTION Select standard or high dosing using local prescribing criteria and renal function.

First-line entries in Table 6

Antimicrobial / option	Dose and route	Course	Selection / conditions
Ciprofloxacin	400 mg IV twice daily	By case	Check local resistance and fluoroquinolone restrictions.
Levofloxacin	Standard: 500 mg PO once daily; high dose: 500 mg PO twice daily	By case	Check local resistance and fluoroquinolone restrictions; adjust for renal function.
Cefotaxime	2 g IV three times daily	By case	EAU notes that monotherapy was not studied in acute pyelonephritis.
Ceftriaxone	Standard: 2 g IV once daily; high dose: 2 g IV twice daily	By case	EAU favours higher dosing than the lower dose studied; select according to patient/context.

Second-line entries in Table 6

Antimicrobial / option	Dose and route	Course	Selection / conditions
Cefepime	Standard: 1 g IV three times daily OR 2 g IV twice daily; high: 2 g IV three times daily	By case	Renal adjustment is important; assess neurotoxicity risk.
Piperacillin-tazobactam	Standard: 4.5 g IV three times daily; high: 4.5 g IV four times daily, prolonged infusion	By case	Prolonged-infusion preparation/timing follows local protocol.
Gentamicin	6-7 mg/kg IV once daily	By case	Not established as pyelonephritis monotherapy in the table; weight, renal function and therapeutic drug monitoring required.
Amikacin	25-30 mg/kg IV once daily	By case	Weight-based specialist/local dosing and therapeutic drug monitoring.

02 / Urological infections

06 Pyelonephritis: resistant-organism options

RESERVE
Resistance-directed options

VERIFY
Component doses + infusion

NARROW
Use susceptibility results

RESERVE USE These are Table 6 last-line entries, not a routine empirical menu. ESBL risk in sepsis is a separate clinical consideration; select with microbiology/infectious-disease input.

VERIFY COMBINATION EAU prints meropenem-vaborbactam as 2 g three times daily, without component clarification. The current EMA product information specifies 2 g meropenem + 2 g vaborbactam every 8 h for CrCl >=40 mL/min. The difference is explicit below.

Table 6 reserve regimens

Antimicrobial / option	Dose and route	Course	Selection / conditions
Imipenem-cilastatin	Standard: 0.5 g IV four times daily over 30 min; high: 1 g IV four times daily over 30 min	By case	EAU dose notation retained; verify formulation/component convention, renal dose and seizure risk.
Meropenem	1 g IV three times daily	By case	Select by susceptibility and local infusion protocol.
Ceftolozane-tazobactam	1.5 g IV three times daily	By case	Combination dose as EAU lists; renal adjustment required.
Ceftazidime-avibactam	2.5 g IV three times daily	By case	Combination dose as EAU lists; susceptibility-directed.
Cefiderocol	2 g IV three times daily	By case	Reserve option; check renal/augmented-clearance dosing and infusion requirements.
Meropenem-vaborbactam - VERIFY	EAU: 2 g three times daily (ambiguous). EMA: 2 g/2 g IV every 8 h, infused over 3 h	EMA: 5-10 days*	EMA dose applies at CrCl >=40 mL/min; lower clearance needs adjustment. *May continue up to 14 days per product information.
Plazomicin	15 mg/kg IV once daily	By case	Check availability, weight/renal dosing and therapeutic drug monitoring.

Not a complete resistant-pathogen formulary

Antimicrobial / option	Dose and route	Course	Selection / conditions
Other resistance-directed agents	Choose with microbiology / infectious diseases	By case	Parenteral fosfomycin is discussed for selected resistant systemic UTI, but EAU 3.7 supplies no universal dose.
Oral switch	Only to a proven active and site-appropriate agent	Total course	Clinical improvement alone does not establish oral-agent susceptibility.

02 / Urological infections

07 Systemic UTI: treatment selection

ASSESS
Severity + resistance risk

TREAT
Empirical then directed

CONTROL
Obstruction / infection source

DECISION Systemic UTI includes pyelonephritis, acute prostatitis and urosepsis. Table entries below are selection pathways; use the dose tables on pp 4-6 only where clinically appropriate.

FLUOROQUINOLONES EAU Strong: avoid empirical ciprofloxacin/other fluoroquinolones in urology-department patients or after fluoroquinolone use within 6 months. Local resistance <10% is necessary but does not remove these exclusions.

PRACTICAL Assess previous isolates, recent antibiotics, ESBL risk, allergy and organ function. Obtain appropriate cultures and reassess the source; broad initial coverage needs a documented narrowing plan.

EAU empirical systemic-UTI pathways

Antimicrobial / option	Dose and route	Course	Selection / conditions
Amoxicillin + aminoglycoside	IV combination; section does not give component doses	See duration below	One of the EAU Strong empirical IV options; use local component doses and monitoring.
Second-generation cephalosporin + aminoglycoside	IV combination; component doses not supplied here	See duration below	EAU Strong empirical IV option.
Third-generation cephalosporin	IV; ceftriaxone/cefotaxime examples on p 5	See duration below	EAU Strong empirical IV option.
Ciprofloxacin (restricted)	Use a site-appropriate dose from pp 4-5	5-7 days*	Only if EAU selection criteria allow: resistance <10%, contraindications to usual alternatives/beta-lactam hypersensitivity; observe exclusions above.
Critical illness / worsening / suspected obstruction	Meropenem 1 g IV every 8 h OR imipenem 500 mg IV every 6 h	By case	EAU narrative broad-spectrum pathway; urgent assessment of obstruction/source.
MRSA coverage when indicated	Add vancomycin; daptomycin or linezolid are alternatives	By case	EAU gives agents, not universal doses here; select/monitor with local specialist protocol.

Directed therapy: response and duration

Antimicrobial / option	Dose and route	Course	Selection / conditions
Improving within 48-72 h	Fluoroquinolone 5-7 days; TMP-SMX 7-10 days; beta-lactam 7-10 days	Total course	*Generic systemic-UTI guidance; syndrome-specific courses take precedence.
Men with systemic UTI	7 days was inferior to 14 days in cited evidence	Consider 14 days	Apply the male febrile-UTI/prostate context; do not default to the shortest generic course.
Slow response / persistent nidus	Investigate and treat source; individualise extension	Reassess at 48-72 h	Do not prolong only because bacteraemia is present; S. aureus bacteraemia needs its own pathway.

02 / Urological infections

08 Catheter-associated UTI and urosepsis

DEVICE
Remove / replace appropriately

CULTURE
Fresh-catheter sample

SEPSIS
Antimicrobials + source control

CA-UTI Treat symptomatic infection according to localised/systemic-UTI pathways. Colonisation, cloudy urine or pyuria alone is not a reason for a universal antibiotic course.

SOURCE STATUS EAU 2026 section 3.9 is under review and directs readers to Surviving Sepsis Campaign 2021. There is no current EAU 2026 urosepsis dose table to reproduce; the short sepsis box below is explicitly supplementary SSC guidance.

CA-UTI: agent and course selection

Antimicrobial / option	Dose and route	Course	Selection / conditions
Localised / systemic active regimen	Select from pp 2 or 4-7 according to site, severity and culture	7 days	For prompt symptom resolution; no universal catheter-specific antibiotic.
Delayed response	Active site-appropriate regimen	14 days	EAU CA-UTI narrative, irrespective of continued catheterisation.
Levofloxacin option	CA-UTI section does not state a separate dose	5 days	May be considered if not severely ill; fluoroquinolone resistance/restrictions still apply.
Selected women <=65 years	Agent-specific active regimen	3 days may be considered	Only after catheter removal, without upper-tract symptoms.
Catheter-associated ABU	No routine antibiotic	None	Exception: before traumatic urinary-tract intervention.
Routine catheter prevention	No prophylactic antimicrobial	None	Do not routinely give prophylaxis for catheterisation, catheter removal or intermittent self-catheterisation.

Device action / supplementary sepsis context

Antimicrobial / option	Dose and route	Course	Selection / conditions
Indwelling catheter	Remove or replace before treatment; culture from newly placed catheter	At assessment	EAU Strong. A catheter present >=2 weeks and still needed should be replaced in symptomatic CA-UTI.
Possible shock / high likelihood of sepsis	Immediate antimicrobials, ideally within 1 hour	SSC 2021	Choose the empirical regimen through local sepsis and systemic-UTI protocols; no single drug suits every case.
Possible sepsis without shock	Rapid investigation; antibiotics within 3 h if concern persists	SSC 2021	Time-limited assessment of infection and non-infectious causes.
Obstruction / abscess / infected device	Urgent source-control assessment + appropriate antimicrobial	SSC / EAU context	Drainage/debridement/device management can be essential; antibiotic escalation alone may be insufficient.

02 / Urological infections

09 Urethritis: empirical and gonococcal therapy

SAMPLE
NAAT + gonococcal culture

TARGET
Likely / proven organism

CONTACTS
Testing + partner care

CONTEXT Assess objective urethritis, exposure, symptoms and follow-up reliability before empirical therapy. Alternatives below are source-listed options, often specialist choices with susceptibility/availability constraints.

VERIFY GC For proven gonorrhoea, the narrative recommends ceftriaxone alone; the pathogen table also prints doxycycline. Doxycycline addresses concurrent/unexcluded chlamydial infection, not gonococcal resistance. Keep the treatment target explicit.

Empirical urethritis

Antimicrobial / option	Dose and route	Course	Selection / conditions
Suspected gonorrhoea	Ceftriaxone 1-2 g IM or IV once PLUS doxycycline 100 mg PO twice daily	Doxy: 7 days	Treat both suspected gonorrhoea and chlamydial infection.
Doxycycline contraindicated / follow-up uncertain	Ceftriaxone as above PLUS azithromycin PO: day 1, 1 g; days 2-4, 500 mg daily	4 days	Alternative empirical combination described in EAU.
Suspected non-gonococcal urethritis	Doxycycline 100 mg PO twice daily	7 days	Empirical first choice.
NGU: doxycycline contraindicated	Azithromycin PO: day 1, 1 g; days 2-4, 500 mg daily	4 days	Alternative when treatment cannot await testing/follow-up.

N. gonorrhoeae detected: main and source-listed alternatives

Antimicrobial / option	Dose and route	Course	Selection / conditions
Ceftriaxone	1-2 g IM or IV once	Single dose	First-line narrative; address chlamydial coinfection separately.
Azithromycin (conditional)	1 g PO once if M. genitalium excluded; otherwise day 1, 1 g then 500 mg/day on days 2-4	1 or 4 days	Narrative: cephalosporin contraindication AND culture-confirmed azithromycin susceptibility.
Ciprofloxacin (conditional)	500 mg PO once	Single dose	Narrative: cephalosporin contraindication AND molecular/culture-confirmed ciprofloxacin susceptibility.
Cefixime + azithromycin	400 mg PO once + 1 g PO once	Single doses	Source-listed alternative; specialist/local resistance guidance.
Gentamicin + azithromycin	240 mg IM once + 2 g PO once	Single doses	Source-listed alternative.
Gemifloxacin + azithromycin	320 mg PO once + 2 g PO once	Single doses	Source-listed alternative; check availability.
Spectinomycin	2 g IM once	Single dose	Source-listed alternative; check infection site and availability.
Fosfomycin trometamol	3 g PO on days 1, 3 and 5	3 doses	Source-listed alternative, not routine first-line gonorrhoea therapy.

02 / Urological infections

10 Urethritis: pathogen-directed non-GC therapy

IDENTIFY
Pathogen + resistance

SELECT
Organism-specific course

FOLLOW UP
Symptoms / contacts / cure test

SPECIES M. hominis and U. parvum detection generally represents colonisation and does not warrant treatment. Treat U. urealyticum only when it is the sole plausible cause of symptomatic urethritis.

M. GENITALIUM Use resistance results when available. A repeat empirical macrolide course after failure is not an automatic next step; EAU provides a resistance/failure-based pathway.

Organism / antimicrobial regimens

Antimicrobial / option	Dose and route	Course	Selection / conditions
C. trachomatis: doxycycline	100 mg PO twice daily	7 days	First-line.
C. trachomatis: azithromycin	1 g PO once if M. genitalium excluded; otherwise 1 g day 1, then 500 mg daily days 2-4	1 or 4 days	Alternative when doxycycline cannot be used.
C. trachomatis: levofloxacin	500 mg PO once daily	7 days	Selected second-line cases when other agents cannot be used.
C. trachomatis: ofloxacin	200 mg PO twice daily	7 days	Selected second-line cases.
M. genitalium: azithromycin	1 g PO day 1, then 500 mg PO daily days 2-4	4 days	EAU pathway when macrolide resistance is not identified; assess local resistance context.
M. genitalium: moxifloxacin	400 mg PO once daily	7 days	Macrolide resistance/high resistance likelihood, azithromycin contraindication or failure.
M. genitalium: sitafloxacin	100 mg PO twice daily	7 days	Narrative rescue option if azithromycin and moxifloxacin fail/cannot be used; specialist and availability check.
U. urealyticum: doxycycline	100 mg PO twice daily	7 days	Only clinically causal symptomatic infection.
U. urealyticum: azithromycin	1 g PO once	Single dose	Alternative if doxycycline contraindicated.
T. vaginalis: metronidazole	1.5-2 g PO once	Single dose	Urethritis regimen; do not substitute for the prostatitis course.
T. vaginalis: tinidazole	2 g PO once	Single dose	Alternative if metronidazole cannot be used.

Follow-up tasks (not additional drug courses)

Antimicrobial / option	Dose and route	Course	Selection / conditions
Sexual contact / partner care	Avoid sex for >=1 week after completing therapy, longer if symptoms persist	EAU narrative	Assess/test/treat relevant partners; offer other STI/HIV testing.
Persistent / recurrent symptoms	Reassess organism, adherence, reinfection and resistance	After >2 weeks	EAU schedules cure testing at 6-12 weeks for initial GC/CT/M. genitalium; coordinate pathogen-specific local STI guidance.

02 / Urological infections

11 Acute and chronic bacterial prostatitis

ACUTE
Systemic-UTI pathway

CHRONIC
Culture + tissue penetration

VERIFY
Conflicting doxycycline duration

ACUTE Do not perform prostatic massage in suspected acute bacterial prostatitis. Consider retention and abscess. Antimicrobial therapy must be paired with drainage/source management when indicated.

CBP Chronic pelvic pain without demonstrated bacterial infection is not automatically chronic bacterial prostatitis. Repeated antibiotics require a defensible infection diagnosis.

SOURCE CONFLICT EAU prostatitis Table 10 prints doxycycline for 10 days; the narrative says doxycycline/azithromycin should extend to at least 3-4 weeks. Both statements are flagged below; resolve the intended course before prescribing.

Acute bacterial prostatitis (ABP)

Antimicrobial / option	Dose and route	Course	Selection / conditions
Active systemic-UTI IV regimen	Third-generation cephalosporin, broad-spectrum penicillin or fluoroquinolone; may combine with aminoglycoside	2-4 weeks total	EAU Strong directs ABP to systemic-UTI recommendations. Use pp 5-7 for contextual doses; severe illness favours IV therapy.
Oral step-down	Susceptible, prostate-appropriate agent after improvement	Complete total course	The section supplies no single universal oral drug/dose; select by culture and clinical response.

Chronic bacterial prostatitis: explicit source table

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fluoroquinolone (e.g. ciprofloxacin, levofloxacin)	EAU: optimal oral daily dose; numeric dose not specified	4-6 weeks	First-line CBP recommendation (Strong); choose the product-specific dose and susceptibility.
Doxycycline - VERIFY	100 mg PO twice daily	Table: 10 days	Table limits use to C. trachomatis / mycoplasma. Narrative: >=3-4 weeks; unresolved course discrepancy.
Azithromycin - VERIFY	500 mg PO once daily	Table: up to 3 weeks	Table limits use to C. trachomatis. Narrative: >=3-4 weeks; discuss intended course with specialist.
Metronidazole	500 mg PO three times daily	14 days	T. vaginalis-associated CBP only.

Source-control and review points

Antimicrobial / option	Dose and route	Course	Selection / conditions
Urinary retention	Assess urethral/suprapubic drainage	As indicated	No additional antibacterial regimen replaces bladder drainage when needed.
Prostatic abscess	Assess image-guided / surgical drainage	As indicated	Persistent fever/poor response warrants reassessment.
Intraprostatic antibiotic injection	Not a recommended option	None	No controlled evaluation supporting routine use.

02 / Urological infections

12 Acute epididymitis

EXCLUDE
Torsion / abscess

SAMPLE
Urine culture + STI NAAT

COVER
Likely STI / Enterobacterales

ASSESS Age and sexual history alone do not reliably exclude an STI. Use the clinical presentation, urine culture and first-void NAAT to select coverage. Severe illness, fever, abscess or uncontrolled pain may require admission and parenteral treatment.

COURSE The 10-14 day ranges below depend on pathogen identification and clinical response. For gonorrhoea, ceftriaxone is additional to chlamydial coverage.

Empirical antibiotic pathways

Antimicrobial / option	Dose and route	Course	Selection / conditions
Low gonorrhoea risk: single-agent option	Levofloxacin 500 mg PO once daily	10-14 days	Covers likely Enterobacterales and C. trachomatis in this EAU pathway; adapt to susceptibility.
Low gonorrhoea risk: combination option	Doxycycline 200 mg PO initial dose, then 100 mg twice daily PLUS TMP-SMX 160/800 mg PO twice daily	10-14 days	Both agents form one combination regimen, not interchangeable alternatives.
Likely gonorrhoeal epididymitis	Ceftriaxone 1 g IM or IV once PLUS doxycycline 200 mg PO initial dose, then 100 mg twice daily	Ceftriaxone once; doxy 10-14 days	Coverage of both N. gonorrhoeae and C. trachomatis.
Non-sexually active: option A	Levofloxacin 500 mg PO once daily	10-14 days	Targets Enterobacterales; obtain culture and assess urinary-tract risk factors.
Non-sexually active: option B	TMP-SMX 160/800 mg PO twice daily	10-14 days	Alternative Enterobacterales-active regimen if appropriate.
Severe infection / hospitalisation	Parenteral regimen guided by severity, culture and local protocol	By case	EAU does not supply a separate numeric IV epididymitis table.
Suspected TB / fungal / viral cause	Cause-specific management, not routine empirical antibacterial extension	Cause-specific	TB: p 15. Candida is rare; fungal work-up/specialist input. Mumps requires a viral differential.

Clinical review and source control

Antimicrobial / option	Dose and route	Course	Selection / conditions
Drainable abscess / necrotic tissue	Urological exploration/drainage if indicated	Urgent assessment	Antibiotics alone may be insufficient.
STI-associated disease	Partner tracing/treatment; avoid sex until free of infection	Review after therapy	EAU describes STI review 14 days after treatment to confirm clinical cure and partner care.
Supportive measures	Analgesia, local cooling and scrotal elevation	As needed	Supportive care accompanies antimicrobial treatment.

02 / Urological infections

13 Fournier gangrene

EMERGENCY
Immediate broad IV therapy

OPERATE
Early complete debridement

REFINE
Cultures + clinical response

PRIORITY Suspected necrotising perineal infection needs urgent surgical assessment. EAU recommends repeated debridement commencing within 24 h; this is not a reason to wait when earlier surgery is feasible.

COMBINATIONS PLUS means drugs given together. OR means an alternative. Table 11 lists some standalone agents without explaining every combination; the gentamicin entry is explicitly not presented here as adequate monotherapy.

DURATION EAU Table 11 gives doses but no fixed antimicrobial duration. Individualise to debridement/source control, microbiology and clinical response.

Source-listed broad-spectrum regimens

Antimicrobial / option	Dose and route	Course	Selection / conditions
Piperacillin-tazobactam PLUS vancomycin	Pip-tazo 4.5 g IV three or four times daily; vancomycin 15 mg/kg IV twice daily	By case	Source dose facts; vancomycin requires renal/weight adjustment and exposure monitoring.
Imipenem-cilastatin	Standard: 0.5 g IV four times daily over 30 min; high: 1 g IV four times daily over 30 min	By case	MRSA/other adjunct coverage depends on context; verify component convention and renal dosing.
Meropenem	1 g IV three times daily	By case	Broad-spectrum option; MRSA-active coverage may still be needed.
Ertapenem	1 g IV once daily	By case	Listed option; does not provide Pseudomonas or MRSA coverage.
Gentamicin - adjunct entry	6-7 mg/kg IV once daily	By case	Not a complete standalone Fournier regimen; use only in a complete coverage plan with monitoring.
Cefotaxime PLUS metronidazole OR clindamycin	Cefotaxime 2 g IV four times daily PLUS metronidazole 500 mg IV four times daily OR clindamycin 600-900 mg IV three times daily	By case	The second drug is an alternative adjunct: do not read this as all three drugs automatically.
Cefotaxime PLUS fosfomycin PLUS metronidazole	Cefotaxime 2 g IV four times daily + fosfomycin 5 g IV three times daily + metronidazole 500 mg IV four times daily	By case	Fosfomycin here is IV, not oral trometamol. All three are combined.

Non-drug treatment is central

Antimicrobial / option	Dose and route	Course	Selection / conditions
Surgical source control	Debridement; repeat procedures as required	Emergency	Broad antibiotics do not replace surgery.
Microbiology / stewardship	Deep tissue cultures and directed narrowing	Repeated review	Choose the eventual regimen with surgical and infection specialists.

02 / Urological infections

14 Genital HSV and HPV-related warts

HSV
Antiviral treatment

HPV
Lesion-directed treatment

DISTINGUISH
No antibacterial cure

SCOPE These guideline infections require their own treatment tables even though the therapies are not antibiotics. Do not treat viral lesions with an antibacterial course solely because they occur in the genital tract.

HSV Swab an active lesion for type-specific testing. Table 12 gives episodic regimens, not a comprehensive suppressive-treatment or severe/disseminated-HSV protocol.

Genital HSV: Table 12

Antimicrobial / option	Dose and route	Course	Selection / conditions
First episode: aciclovir option A	400 mg PO three times daily	10 days	One of two aciclovir schedules.
First episode: aciclovir option B	200 mg PO five times daily	10 days	Alternative aciclovir schedule.
First episode: valaciclovir	500 mg PO twice daily	10 days	EAU Table 12 regimen.
Recurrence: aciclovir option A	400 mg PO three times daily	5 days	Episodic treatment.
Recurrence: aciclovir option B	800 mg PO twice daily	5 days	Episodic alternative.
Recurrence: aciclovir option C	800 mg PO three times daily	2 days	Episodic alternative.
Recurrence: valaciclovir	500 mg PO twice daily	3 days	Episodic alternative.

Anogenital warts: recommended local therapies (not antibiotics)

Antimicrobial / option	Dose and route	Course	Selection / conditions
Imiquimod 5% cream	Apply to external warts overnight, 3 nights per week	16 weeks	EAU Strong; check product application/washing instructions.
Sinecatechins 10% or 15%	Topical application three times daily	Until clearance; max 16 weeks	EAU Strong; availability and site suitability matter.
Podophyllotoxin	Topical twice daily for 3 days, then 4 days off	4-5 weeks	EAU Strong; formulation/area restrictions follow product information.
Clinician-administered therapy	Cryotherapy; excision; electrosurgery/electrocautery; laser	Procedure-specific	Alternatives selected by lesion/site and expertise.
Other agents discussed in evidence	No standard recommended regimen provided here	Not a prescribing menu	HPV: salicylic acid, 5-fluorouracil, potassium hydroxide. HSV: pritelivir, amenamevir, inosine pranobex are discussed, not supplied as standard Table 12 regimens.

02 / Urological infections

15 Genitourinary tuberculosis

CONFIRM
TB microbiology + resistance

COMBINE
Multidrug treatment

SPECIALIST
TB service + monitoring

STANDARD Newly diagnosed drug-susceptible GUTB uses a 6-month daily combination: 2 months HRZE, then 4 months HR (EAU Strong). These drugs are combined within each phase, not chosen one at a time.

MDR CAUTION The EAU source retains a legacy A-D MDR-TB grouping. It is recorded as a source inventory below, not a modern standalone MDR prescription. Use current TB-programme/WHO resistance-directed guidance (WHO Module 4, 2nd ed., September 2026).

MONITOR Review renal/hepatic function, interactions, visual toxicity, adherence and pyridoxine requirements with the TB service. Obstruction/destruction may require urological intervention.

Drug-susceptible GUTB: daily treatment facts

Antimicrobial / option	Dose and route	Course	Selection / conditions
Isoniazid (H)	5 mg/kg PO once daily; maximum 300 mg/day	Months 1-6	Part of both phases.
Rifampicin (R)	10 mg/kg PO once daily; maximum 600 mg/day	Months 1-6	Part of both phases; major interaction potential.
Pyrazinamide (Z)	25 mg/kg PO once daily; maximum 2,000 mg/day	Months 1-2	Intensive phase only in this regimen.
Ethambutol (E)	15-20 mg/kg PO once daily; EAU maximum 800-1,600 mg/day by weight	Months 1-2	Intensive phase; weight/renal/visual monitoring.

EAU MDR-TB agent inventory - specialist selection only

Antimicrobial / option	Dose and route	Course	Selection / conditions
Group A: fluoroquinolones	Levofloxacin; moxifloxacin; gatifloxacin	Not specified	Legacy source grouping; not a recommendation to use every listed drug.
Group B: injectables	Amikacin; capreomycin; kanamycin; streptomycin	Not specified	Streptomycin substitution only with susceptibility and no usable listed alternative in the source.
Group C	Ethionamide/prothionamide; cycloserine/terizidone; linezolid; clofazimine	Not specified	Numeric doses are not supplied in this EAU table.
Group D1 / D2	D1: pyrazinamide, ethambutol, high-dose isoniazid. D2: bedaquiline, delamanid	Not specified	Do not build an MDR regimen from this inventory alone.
Group D3	PAS; imipenem-cilastatin; meropenem; amoxicillin-clavulanate; thioacetazone	Not specified	Source inventory; thioacetazone must not be used in HIV-positive patients.

02 / Urological infections

16 Localised fungal UTI

SPECIES
Culture + susceptibility

LOCALISE
Bladder vs upper/systemic

VERIFY
Ambiguous instillation entry

NO AUTOMATIC TREATMENT Asymptomatic funguria usually needs no antifungal; exceptions need individual assessment (neutropenia, very-low-birth-weight infants, mucosa-breaching procedures). This reference is adult-focused; it supplies no neonatal regimen.

SOURCE ISSUES The local amphotericin instillation table contains unusable units; that recipe is not reproduced. The narrative supplies a separate 50 mg/L irrigation option. Caspofungin is listed for localised infection despite a narrative warning about poor urinary penetration; use only after specialist review.

SPECIES Do not infer C. glabrata fluconazole susceptibility from species alone. Require testing; fluconazole is inappropriate for a resistant isolate. Remove/replace implicated devices and relieve obstruction where indicated.

Symptomatic localised fungal UTI: regimen facts

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fluconazole: susceptible species	200 mg PO once daily	14 days	Narrative describes a double first-day loading dose. CrCl <50 mL/min: source lists 50% dose reduction; verify maintenance/loading with pharmacy.
Fluconazole: susceptible C. glabrata	800 mg PO once daily	14-day framework	Only confirmed susceptible isolate; high-dose specialist decision and renal adjustment.
Amphotericin B deoxycholate: IV	0.3-0.6 mg/kg IV once daily	1-7 days	Resistant species; with/without flucytosine for C. glabrata. Source states no renal dose adjustment, but nephrotoxicity/electrolyte monitoring remains essential.
Amphotericin B deoxycholate: irrigation - VERIFY	Narrative: continuous bladder irrigation 50 mg/L	5 days	Selected local fluconazole-resistant cases only after excluding upper/systemic spread. Table recipe has invalid units: do not use it as a preparation instruction.
Flucytosine	25 mg/kg PO or IV four times daily	14 days	May be monotherapy for C. glabrata in the source. Verify local formulation/availability, renal dosing, blood counts and therapeutic levels.
Caspofungin - VERIFY	70 mg IV on day 1, then 50 mg IV once daily	2-3 weeks	Source table option; urinary-penetration concern in the same chapter. Not a routine bladder-infection choice.

Renal/dialysis entries from the fungal table

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fluconazole	CrCl <50: 50% dose reduction; haemodialysis: full dose 3 times weekly after dialysis	Verify schedule	This table does not cover all dialysis/CRRT schedules.
Flucytosine	CrCl 21-40: 25 mg/kg twice daily; 10-20: once daily; <10: every 48 h	CrCl in mL/min	Dialysis: source lists 25-50 mg/kg every 48-72 h after dialysis; specialist levels-based dosing required.

02 / Urological infections

17 Systemic fungal UTI

ASSESS
Upper tract / systemic spread

DRAIN
Obstruction / devices

TARGET
Species + susceptibility

SCOPE The source table uses systemic fungal UTI, not a complete candidemia/invasive-candidiasis protocol. Haematogenous infection, candidemia, fungus balls or an obstructed system need specialist assessment and source control.

FORMULATION Amphotericin B here means deoxycholate. Do not substitute a lipid formulation mg-for-mg or assume equal urinary exposure.

RENAL No renal dose reduction in an EAU row does not mean absence of nephrotoxicity. Monitor creatinine, electrolytes and interacting/nephrotoxic drugs; use pharmacy-supported dosing.

Systemic fungal UTI: source regimen facts

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fluconazole: susceptible species	200-400 mg PO once daily	14 days	Severe infection: source considers 400/800 mg loading on day 1 according to chosen maintenance dose.
Fluconazole: susceptible C. glabrata	800 mg PO once daily	14 days	Susceptibility-confirmed high-dose therapy; specialist supervision.
Amphotericin B deoxycholate	0.3-0.6 mg/kg IV once daily	1-7 days	Fluconazole-resistant species; with/without flucytosine for C. glabrata.
Flucytosine	25 mg/kg PO or IV four times daily	14 days	Generally combined with amphotericin B in the systemic table.
Caspofungin	70 mg IV day 1, then 50 mg IV once daily	2-3 weeks	Renal tissue exposure differs from urinary exposure; select according to infection site/species and specialist advice.
Micafungin (narrative only)	No numeric EAU UTI dose supplied	Not specified	Discussed for exceptional fluconazole-resistant cases; not an additional fixed regimen.
Ketoconazole / voriconazole	Small case-series discussion, no routine EAU UTI regimen	Not specified	Do not promote evidence-discussion agents to standard treatment options.

Renal adjustment / source control

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fluconazole	CrCl <50 mL/min: source lists 50% dose reduction	Maintenance check	Dialysis: source lists full dose 3 times weekly after dialysis; loading/CRRT need separate review.
Flucytosine	CrCl 21-40: 25 mg/kg twice daily; 10-20: daily; <10: every 48 h	Renal adjustment	Dialysis: 25-50 mg/kg every 48-72 h after dialysis in source; verify using drug levels.
Caspofungin / amphotericin B	No renal dose adjustment in EAU table	Still monitor	Hepatic dosing and toxicity checks remain agent-specific.
Obstruction / foreign material	Drain obstructed system; assess catheter/stent replacement	Source control	Antifungal selection does not remove the need to manage the cause.

02 / Urological infections

18 Periprocedural antimicrobial prophylaxis

PREVENTION
Not infection treatment

CULTURE
ABU before mucosal breach

LOCALISE
Procedure + local resistance

TABLE STATUS EAU Table 15 lists possible agents, not one mandatory drug for each procedure. A blank drug/dose field below means the source does not supply it; do not fill it from memory.

BIOPSY Prefer transperineal biopsy (EAU Strong). Omit prophylaxis for transperineal biopsy without infection risk factors (Weak). Fluoroquinolones are not recommended for prostate-biopsy prophylaxis (Strong).

FOSFOMYCIN The prostate-biopsy indication has been withdrawn in Germany. Check national authorisation/local guidance before considering this source-listed option.

Procedure-specific prevention

Antimicrobial / option	Dose and route	Course	Selection / conditions
Urodynamics / cystoscopy	No routine antibiotic prophylaxis	None	EAU recommendations; assess any separate active-infection indication.
Shock-wave lithotripsy	No routine prophylaxis	None	Use antibiotics if bacteriuria/suspected or diagnosed infected stone indicates them.
Ureteroscopy	Possible choices: trimethoprim; TMP-SMX; group 2/3 cephalosporin; aminopenicillin + beta-lactamase inhibitor	Not fixed here	Prophylaxis recommended; Table 15 does not fix dose/course.
Percutaneous nephrolithotomy	Agent chosen using local susceptibility data	Single dose	EAU Strong single-dose prevention recommendation; no fixed drug/dose in Table 15.
TURP	Local, culture-informed prophylactic agent	Not fixed here	Prophylaxis recommended; source does not nominate a universal dose.
TURBT	Local prophylactic agent in high-risk patients	Not fixed here	For high postoperative-sepsis risk, not automatic for every patient.
Transrectal prostate biopsy	First: rectal/stool-culture targeted; second: augmented prophylaxis	Protocol-specific	Rectal povidone-iodine cleansing also recommended. Augmented means >=2 antimicrobial classes.

Transrectal-biopsy examples in Table 15 (possible alternatives)

Antimicrobial / option	Dose and route	Course	Selection / conditions
Fosfomycin trometamol	3 g PO before biopsy (source example)	Verify timing	National authorisation warning above; not a universal approved regimen.
Ceftriaxone	1 g IM	Schedule not fixed	Example in source; follow the chosen local prevention protocol.
Cefixime	400 mg PO; source gives no frequency	3 days, starting 24 h before	VERIFY frequency before use; do not infer once-daily dosing from this text.
Gentamicin / amikacin	Gentamicin 6-7 mg/kg IV once daily OR amikacin 25-30 mg/kg IV once daily	Protocol-specific	Weight, renal function, monitoring and local prophylaxis timing required.